

9. (a) The Earth's early atmosphere around 4 000 million years ago contained mainly carbon dioxide and water vapour produced by volcanoes.

- (i) Explain why the large percentage of water vapour in the Earth's atmosphere decreased over geological time. [2]

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- (ii) Give **two** reasons why the percentage of carbon dioxide in the Earth's atmosphere has decreased over geological time. [2]

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- (b) During the last 250 years the percentage of carbon dioxide in the Earth's atmosphere has increased from 0.03 % to 0.04 %. This has led to increased global warming. Give **one** reason for this increase and explain why global warming is a cause for concern. [2]

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- (c) Ammonia present in the Earth's early atmosphere reacted with oxygen to produce nitrogen and water vapour. Complete the balancing of the symbol equation for this reaction. [1]



END OF PAPER

7



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



24

1	H	
Hydrogen	1	

7	Li	9	Be
Lithium	3	Beryllium	4
23	Na	24	Mg
Sodium	11	Magnesium	12
39	K	40	Ca
Potassium	19	Calcium	20
86	Rb	88	Sr
Rubidium	37	Strontium	38
133	Cs	137	Ba
Caesium	55	Barium	56
223	Fr	226	Ra
Francium	87	Radium	88
		227	Ac
		Actinium	89

11	B	12	C	14	N	16	O	19	F	20	Ne
Boron	5	Carbon	6	Nitrogen	7	Oxygen	8	Fluorine	9	Neon	10
27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar
Aluminium	13	Silicon	14	Phosphorus	15	Sulfur	16	Chlorine	17	Argon	18
70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
Gallium	31	Germanium	32	Arsenic	33	Selenium	34	Bromine	35	Krypton	36
115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
Indium	49	Tin	50	Antimony	51	Tellurium	52	Iodine	53	Xenon	54
204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
Thallium	81	Lead	82	Bismuth	83	Polonium	84	Astatine	85	Radon	86

59	Co	56	Fe	55	Mn	52	Cr	51	V	48	Ti	45	Sc
Cobalt	27	Iron	26	Manganese	25	Chromium	24	Vanadium	23	Titanium	22	Scandium	21
103	Rh	101	Ru	99	Tc	96	Mo	93	Nb	91	Zr	89	Y
Rhodium	45	Ruthenium	44	Technetium	43	Molybdenum	42	Niobium	41	Zirconium	40	Yttrium	39
192	Ir	190	Os	186	Re	184	W	181	Ta	179	Hf	139	La
Iridium	77	Osmium	76	Rhenium	75	Tungsten	74	Tantalum	73	Hafnium	72	Lanthanum	57
195	Pt	197	Au	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As
Platinum	78	Gold	79	Copper	29	Zinc	30	Gallium	31	Germanium	32	Arsenic	33
106	Pd	108	Ag	112	Cd	115	In	119	Sn	122	Sb	128	Te
Palladium	46	Silver	47	Cadmium	48	Indium	49	Tin	50	Antimony	51	Tellurium	52

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number